The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

October 26, 2026

1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 Distributed Multi-Directory Agent Initializations

Each independent neuro-symbolic agent $a_k \in \mathcal{A}$ maintains its unique background logic constraints inside a dedicated disk directory path:

$$a_k \equiv \langle \mathcal{V}_k, \mathcal{K}_k, \mathcal{F}_k \rangle \tag{1}$$

3 Logical Soundness of the Inductively Grown Merger Ontology

We formalize the deductive guarantees that preserve structural sanity and prevent concept drift during the inductive tree retraining pass.

3.1 The Prolog Guardrail Attention Filter Invariant

Let $\mathcal{K}_{\text{target}}$ represent the target agent's verified background knowledge base (`kb.pl`), and let $\Delta\mathcal{E}$ represent the set of repaired, newly aligned complex exemplars staged during the single-pass blind interrogation phase. The inductive learning engine $\mathcal{I}_{\text{CART}}$ generates the revised decision tree structure $\mathcal{T}_{\text{merged}}$ strictly within the declarative perimeter enforced by the logic filter:

$$\mathcal{T}_{\text{merged}} = \mathcal{I}_{\text{CART}}(\mathcal{E} \cup \Delta\mathcal{E}) \quad \text{subject to} \quad \mathcal{K}_{\text{target}} \vdash \Delta\mathcal{E} \nvdash \bot \tag{2}$$

Because the logic verification gateway blocks any exemplar mutation that violates the core structural invariants ($\mathcal{K} \cup \Delta\mathcal{E} \vdash \bot$), the tree is restricted from generating illegitimate feature splits or chasing noise.

3.2 Inductive Definability and Semantic Consistency

The resulting decision forest is completely inductively definable and remains a sound semantic extension of the initial logic. Any path tracing from the root vertex to a newly merged classification leaf matches a validated theorem path within the underlying Description Logic:

$$\forall \text{Path}(\text{Root} \to \mathcal{E}_{\text{leaf}}) \implies \mathcal{K}_{\text{target}} \vdash \Psi_{\text{consistent}} \tag{3}$$

This dual-layer constraint system guarantees that while the agent's ontology expands dynamically to absorb shared multi-scale grid strategies, the operational execution plane retains absolute semantic soundness and boundary integrity with zero black-box VRAM overhead.